



Cambridge International AS & A Level

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PHYSICS

9702/32

Paper 3 Advanced Practical Skills 2

May/June 2026

2 hours

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
1	
2	
Total	

This document has **16** pages. Any blank pages are indicated.

You may not need to use all of the materials provided.

1 In this experiment, you will investigate the equilibrium of forces.

- (a)
- Assemble the apparatus as shown in Figure 1.1.
 - Adjust the bosses so that both nails are 55.0 cm above the bench.
 - Hang a mass D of 300 g from the loop of the short string.
 - Adjust the distance between the stands so that the section of string between points A and B is perpendicular to the section of string between points B and C.

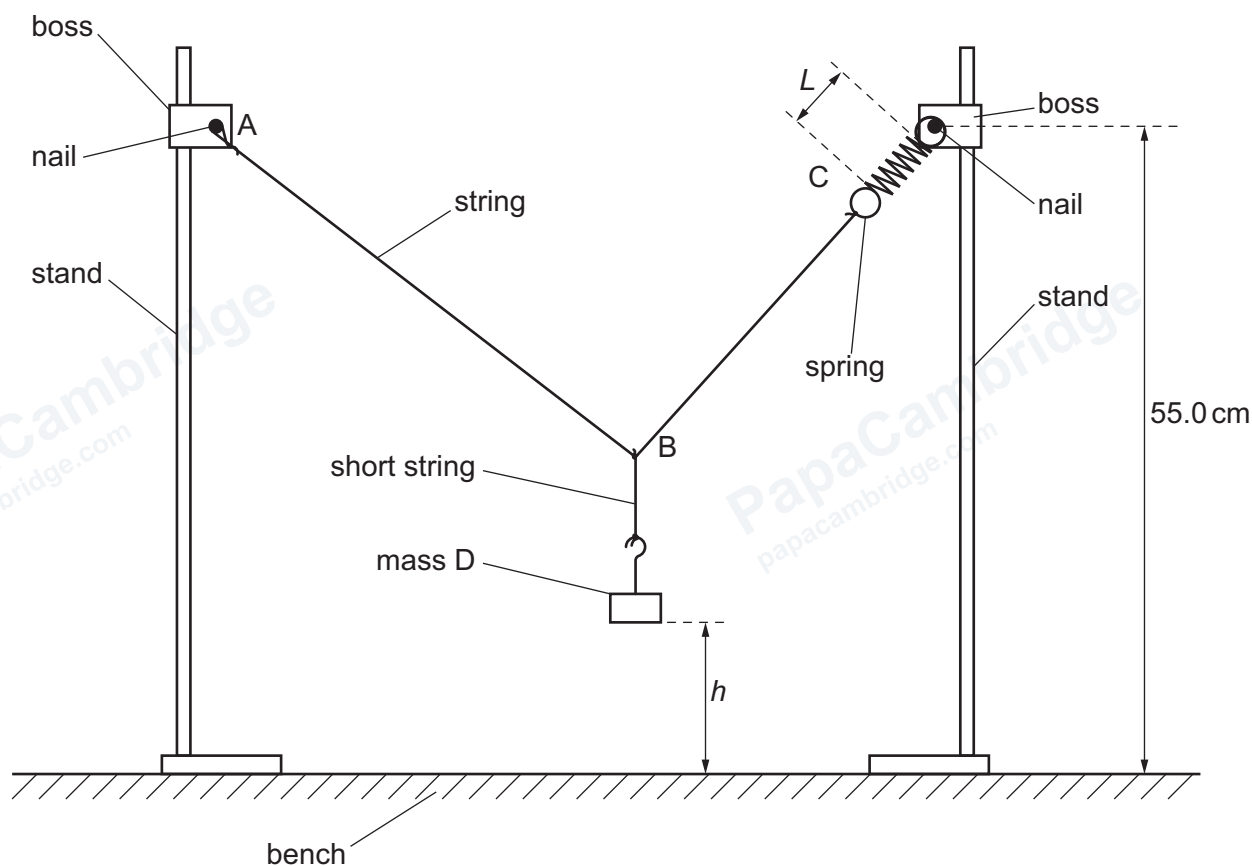


Figure 1.1

- The mass of D is m . Record m .

$m =$

- The length of the coiled part of the spring is L .

The distance between the bottom of D and the bench is h .

Measure and record L and h .

$L =$

$h =$

- Calculate T , where $T = L + h$.

$T =$

[2]



- (b) Change m and adjust the distance between the stands so that the section of string between points A and B is perpendicular to the section of string between points B and C. Measure and record L and h . Repeat until you have six sets of values of m , L and h .

Record your results in a table. Include values of T in your table.

[10]

- (c) (i) Plot a graph of T on the y -axis against m on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

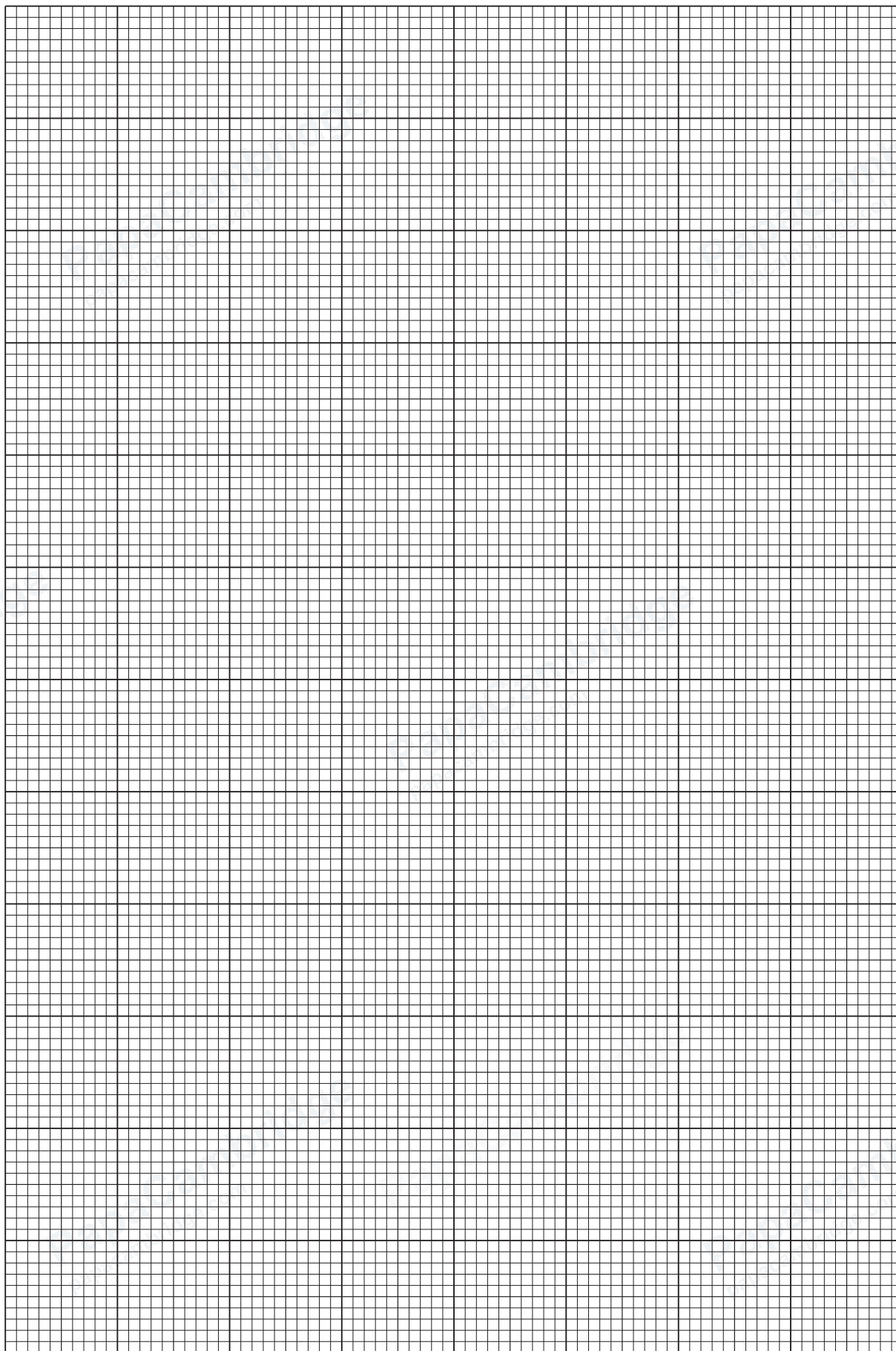
- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]





(d) It is suggested that the quantities T and m are related by the equation

$$T = pm + q$$

where p and q are constants.

Use your answers in **1(c)(iii)** to determine the values of p and q .
Give appropriate units.

$p =$

$q =$

[2]

[Total: 20]

You may not need to use all of the materials provided.

- 2 In this experiment, you will investigate the force applied by water on a copper wire loop.
- (a) You have been provided with three copper wires. Each wire has a hook at one end. Two of the wires have a loop at the other end. The loops have different diameters.
- Using the wire with the loop of the **smaller** diameter, assemble the apparatus as shown in Figure 2.1.
 - The pin passes through the central hole in the card and is pushed into the cork, which is held in the clamp.

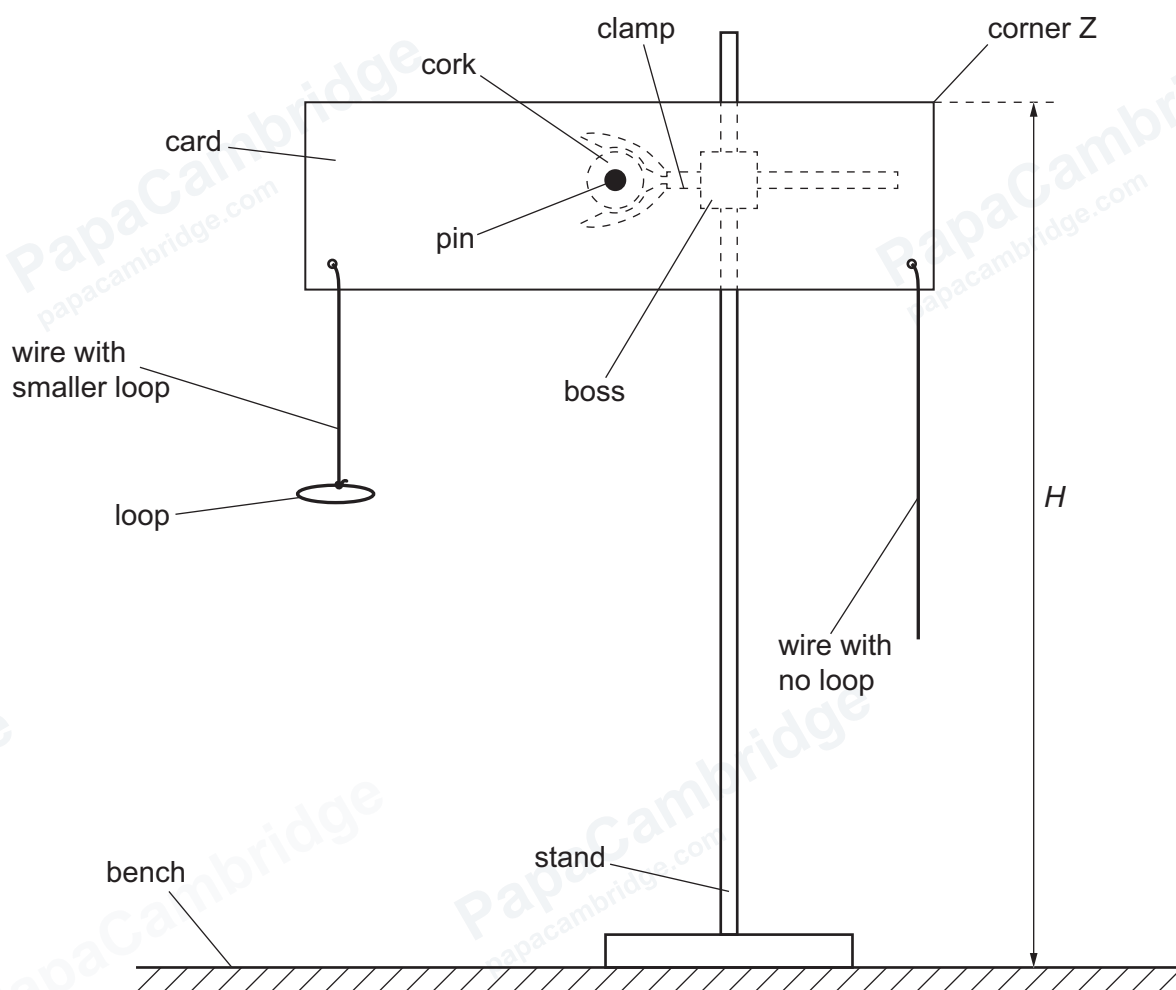


Figure 2.1

- Adjust the height of the boss holding the clamp so that the pin is approximately 40 cm above the bench.
- Check that the card pivots freely on the pin.

- The height of corner Z above the bench is H , as shown in Figure 2.1.

Measure and record H .

$$H = \dots\dots\dots \text{cm}$$

- In Table 2.1, record this value of H in the row $n = 0$.

[1]

- (b) • You have been provided with four paper clips.

Attach **one** paper clip onto the loop of the wire hanging from the card. The card will tilt.

- The number of paper clips attached onto the loop of the wire is n .

Measure H and record this value in Table 2.1.

Table 2.1

n	H/cm
0	
1	
2	
3	
4	

- Add more paper clips, and record the corresponding values of H in Table 2.1.
- Remove the paper clips from the loop.
- On Figure 2.2, plot a graph of H on the y -axis against n on the x -axis.
- Draw the curved line of best fit.



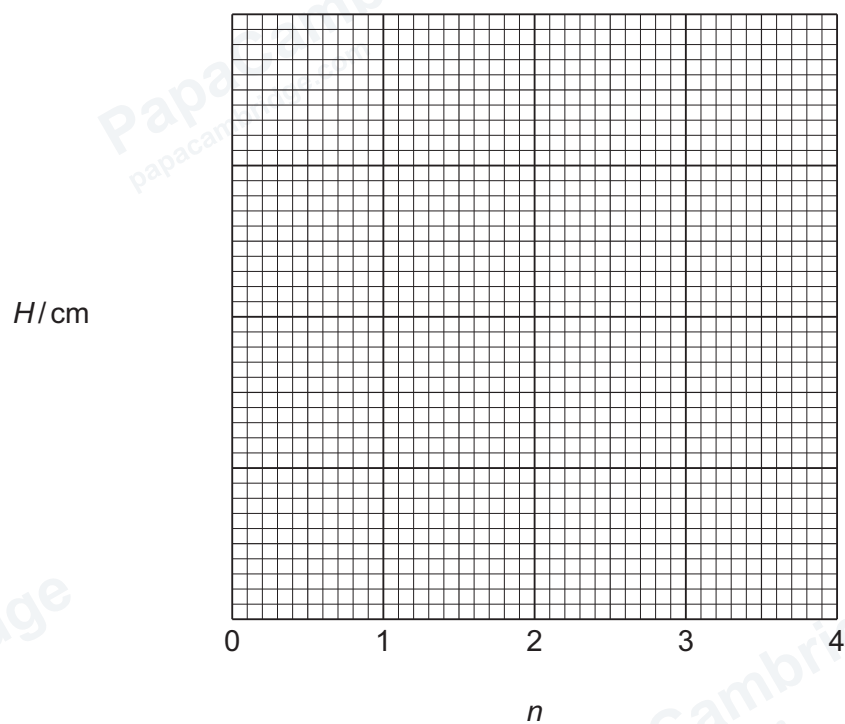


Figure 2.2

[1]

- (c) (i) • Lift up the beaker of water so that the loop just touches the water surface.
- Slowly lower the beaker so that the card tilts, as shown in Figure 2.3.

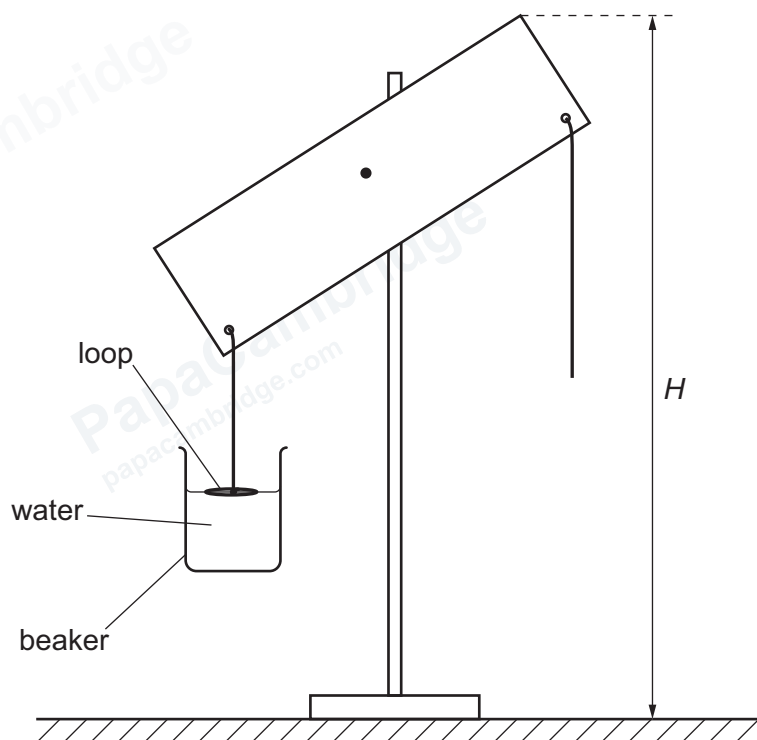


Figure 2.3

- Continue to lower the beaker until the loop breaks away from the water surface.
- Measure and record the value of H at the instant when the loop breaks away from the water surface.

$H = \dots\dots\dots$ cm [2]

- (ii) Estimate the percentage uncertainty in this value of H . Show your working.

percentage uncertainty = $\dots\dots\dots$ % [1]

- (iii) • Use Figure 2.2 to determine the value of n , to the nearest 0.1, that corresponds to this value of H .

$n = \dots\dots\dots$

- The force F applied by the water is given by

$$F = nW$$

where W has the value 0.0032 N.

Calculate F .

$F = \dots\dots\dots$ N [1]

- (d) • Remove the wire with the loop from the card.
- Measure and record the diameter d of the loop.

$d = \dots\dots\dots$ [1]

- (e) • Hang the wire with the **larger** loop from the card.
- Repeat **2(c)(i)**, **2(c)(iii)** and **2(d)** using the wire with the larger loop.

$H =$ cm

$n =$

$F =$ N

$d =$ [2]

- (f) It is suggested that F and d are related by

$$F = 2\pi dk$$

where k is a constant.

- (i) Using your data, calculate **two** values of k .

first value of $k =$

second value of $k =$ [1]

- (ii) Justify the number of significant figures that you have given for your values of k .

.....

.....

..... [1]



(g) It is suggested that the uncertainty in the values of k is 10%.

Using this uncertainty, explain whether your results support the relationship in 2(f).

.....

.....

.....

..... [1]





(h) (i) Describe **four** sources of uncertainty or limitations of the procedure for this experiment.

For any uncertainties in measurement that you describe, you should state the quantity being measured and a reason for the uncertainty.

1

.....

2

.....

3

.....

4

.....

[4]

(ii) Describe **four** improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1

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2

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3

.....

4

.....

[4]

[Total: 20]





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